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Comparative study of Mumbai – Pune and Bengaluru – Tamil Nadu industrial region

AADIL MUBARAK P N

Abstract

The study compares India's most important industrial region such as Mumbai – Pune industrial belt and Bengaluru – Tamil Nadu industrial cluster. This study evaluates both industrial locations historically expansion, infrastructure, and government policy. Industrial cluster when new companies were started or built on a region that have shared infrastructure, geographical advantage, through this, it strengthens the regional production and economic performance. The study mainly depends on the secondary data from the government reports, publications including annual industrial survey (ASI), economic survey and local documents. These two regions were compared across historical background, infrastructure, expansion, employment opportunities, government support and policy. The Mumbai-Pune region has a long industrial history back to the British colonial era, when port access and availability of black soil supported cotton textile industry that later transformed toward heavy industries. Coming to the southern region, it developed more after India got independence, Bengaluru become well known for IT and technology, aerospace, R&D related fields. Chennai developed to global automobile and electronic production. However, both regions face some limitation to grow such as high land cost, environmental pollution, traffic congestion, rapid urban growth. So even if both grew differently both of them are still very important for India's economy today.

Key words

Location of industries, regional development, industrial clusters, comparative study, economic development

1. Introduction

The present paper studies the distribution and economic importance of the industrial clusters in India understanding how and why industries were situated in the same belt or zone that helps to increase the local area and national level economic status (Ragothaman, 2024). Indian economy was dependent on the industrial region like Mumbai-Pune and Bangaluru - Tamil Nadu, which developed due to the environmental factors such as resources and government intervene (Garg, 2019).

This industrial region influenced Indian economy by creating more job opportunities, foreign direct investment (FDI) (Bijarniya L. S., 2025). These industrial clusters increased the manufacturing products, exporting to outer markets, however concentrating all industries in a specific area or region creates imbalance, and how tier-2 and tier-3 regions help fill this gap to reduce regional imbalance (Bijarniya L. S., 2025).

The main objective of this study paper to find out the industrial growth in a region and how they effect the development of neighbouring cities and reduce regional unbalanced development (Murthy, 2025). By studying the locational and other geographical advantages, social and economic impact and government involvement in creating these clusters and also this paper finds that old industrial belts will remain the backbone of Indian manufacturing and new planned development connecting these industrial regions to urban areas, such strategies can promote sustainable, inclusive and regionally balanced economic development in the future (Bijarniya L. S., 2025).

2. Objectives and Research question

- To study how geographical features and state level and national level policies influence foreign direct investment and how it impacts on the employment opportunities and production.
- To study these two industrial regions, help in promoting growth in industries and reducing regional imbalance.

Research Question

How do India's most important industrial clusters lead to regional economic development and create employment opportunities in these sectors, by distinguishing the pattern and investment of both clusters?

3. Database and Methodology

The data taken to compare the Mumbai – Pune and Bengaluru – Tamil Nadu industrial regions in this paper are collected from a trusted source.

3.1 Type of Data

The study uses a mixed dataset, including quantitative data like employment, gross value added(GVA), FDI inflow and manufacturing output, along with qualitative information on cluster history, state industrial policies, and locational advantage. (Tirkey, A comparative analysis: How geographic location and state level policies influence the inflow and impact of FDI in different Indian cities, 2026)

3.2 Sources of Data

The information has been collected from various authentic secondary sources.

3.2.1 Government Reports on Industries

Research papers from Department for Promotion of Industry and Internal Trade (DPIIT), National Institution for Transforming India (NITI Aayog), and the Ministry of Commerce and Industry. (Murthy, 2025)

3.2.2 Industry Statistics

Data sets from the Annual Survey of Industries (ASI) and Periodic Labor Force Survey (PLFS) authorized by the Ministry of Statistics and Programme Implementation

(MoSPI) which provide factory level production and capital parameters. (Ketels, 2023) (Ministry of Statistics and Programme Implementation (MoSPI), 2025)

3.2.3 Official Economic Surveys

Economic survey of India and state-wise economic survey. (Mahatekar, State-wise foreign direct investment (FDI) flows and economic impacts in India: A comparative study (2019–2024), 2024)

3.2.4 Academic Articles, Books, And Industry Studies

Peer reviewed journals, policy literature, and research reports explaining spatial agglomeration, new economic geography and regional development (Majeed, 2022)

3.2.5 Data from Trusted Online Sources

Government of India Open Data Portal (data.gov.in), Reserve Bank of India (RBI), Statistical publications, and census of India urban agglomeration data. (Mahatekar, State-wise foreign direct investment (FDI) flows and economic impacts in India: A comparative study (2019–2024), 2025)

3.3 Time Period

The data covers the time period during economic liberalization in India to recent FDI and ASI reports, specifically from 2010 to 2024

3.4 Methodology

3.4.1 Research Approach

This study uses a mixed method of approach, combining both quantitative and qualitative method. Quantitative data include numbers such as FDI inflow, employment growth, and GVA, which help point out development in specific regions. (e.g. automobile sector in Pune or IT in Bengaluru)

Qualitative data include analysis of state policies and industrial development patterns, which explain the reasons behind observed trends.

3.4.2 Data Collection

The research relies entirely on secondary sources. There is no primary surveys or interviews were conducted through these studies. Secondary data means data and information been taken from that already published or conducted by the organization.

3.4.3 Data Analysis Method

The two regions were compared using several analytical techniques,

- Comparison of statistical indicator – evaluating key economic measure such as Foreign Direct Investment (FDI), equity inflow, total employment, fixed capital investment, and gross value added (GVA).
- Use of tables– to show how regional development on basis of geographical locational advantages, government policies, and economic impact.

3.4.4 Rationale

To compare large economic sectors, consistent state level data from the government sources are required. It was not easy to gather primary data of industrial location due to the vast spread it out, productions, workers data due to the lack of resource and time therefore secondary data were used.
Depending on the trusted source like Annual Survey of India (ASI) will help to authentic and genuine comparison between these regions.

3.4.5 Limitation
A key limitation of using secondary data is many datasets are not updated regularly, leading to time gaps between surveys and detailed industrial data such as ASI are often available only at the state level rather than for a specific city or district.
To overcome this limitation that using both state economic data with local government policy and cluster studies.

4. Study Area

4.1 Geographic Location
This study covers India's two main industrial clusters, such as Mumbai – Pune industrial region and Bengaluru – Tamil Nadu industrial region.

4.1.1 Mumbai–Pune Industrial Region

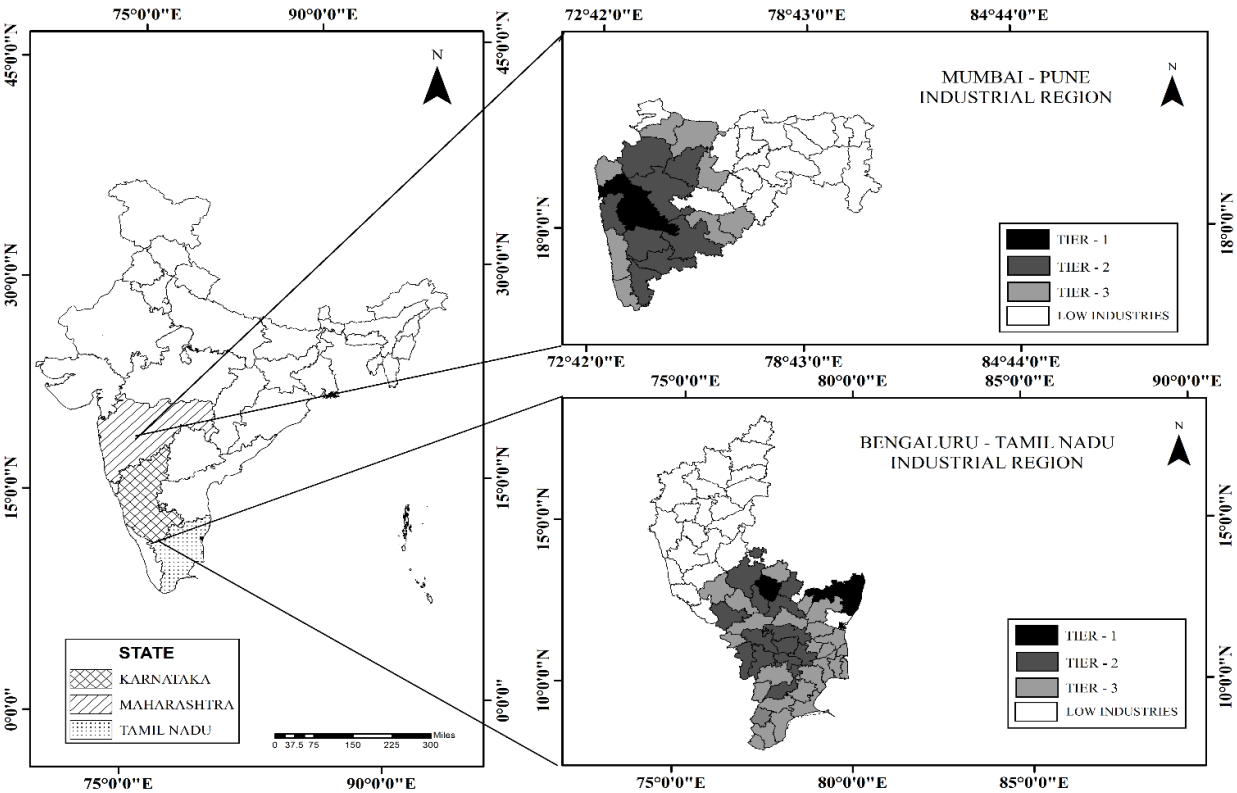

Figure 4.1 industrial regions of Mumbai – Pune and Bengaluru – Tamil Nadu

Mumbai industrial region was situated in the state of Maharashtra in south west of India, having the Arabian Sea to the west extending from Mumbai and Thane to Pune, surrounded by districts like Nashik, Solapur, Colaba, Ahmednagar, Satara, Sangli and Jalgaon. (Garg, 2019) Two metropolitan cities are connected by the 93 kilometres Mumbai – Pune express way. (Ramachandra, 2019).

4.1.2 Bengaluru–Tamil Nadu Industrial Region

Bengaluru – Tamil Nadu industrial region also called Bengaluru – Chennai industrial region that covers the southern state of India which includes Karnataka and Tamil Nadu (Garg, 2019). This industrial belt covers from Bengaluru in Northern Karnataka to Madurai southern Tamil Nadu, that's covering almost all the districts of Tamil Nadu except Villupuram and some southern districts of Karnataka (Vision IAS, 2024).These two states are connected through 560-kilometer-long Chennai – Bengaluru industrial corridor (CBIC) (Japan International Cooperation Agency; PricewaterhouseCoopers Co., Ltd.; Nippon Koei Co., Ltd., 2015) This region also includes the emerging industrial town like Hosur in Tamil Nadu which was developed as electric vehicle and electronics manufacturing hub. (CBRE Research, 2025)

4.2 Physical Features

Physical and climatic features play an important role in the development of these two industrial regions

4.2.1 Mumbai–Pune

The western side, industrial cluster benefits from both physical features such as Jawaharlal Nehru port trust (JNPT) and humid climatic feature which favoured the establishment of classic industries for the cotton and textiles (Vision IAS, 2024) The eastern side is situated near to the Western Ghat hills which helps produce hydroelectric power. (Singh, 2025)

4.2.2 Bengaluru–Tamil Nadu

This industrial cluster has diverse landforms and topography where physically Bengaluru is situated on the Deccan Plateau with an elevation up to 900 meters, plain terrain in the east and south region. Climatically it has moderate temperatures that encourage productivity (Vision IAS, 2024).Chennai is situated on flat coastal terrain stretching 22,800 sq. Km Coromandel Coast of the Bay of Bengal (Hossain, 2022). This region depends on hydroelectric power from local dams like Pykara, Mettur, Sivasamudhram all fed primarily by the river Cauvery (Garg, 2019)

4.3 Economic and Industrial Background

4.3.1 Mumbai–Pune

This is one of the oldest industrial regions, The Mumbai port was opened in 1774 during the British colonial era (Garg, 2019) It was also known as the Manchester of the East for its flourishing cotton mills (Industrial clusters in India: Mumbai–Pune belt

& Bengaluru IT hub, 2024) this region was mainly dominated by the automobiles such as Tata Motors, Mercedes Benz, Volkswagen, petrochemicals, engineering, financial service and IT. (Industrial clusters in India: Mumbai–Pune belt & Bengaluru IT hub, 2024)

4.3.2 Bengaluru–Tamil Nadu

According to (Garg, 2019), the Bengaluru – Tamil Nadu industrial region developed mainly after India gained independence. This region had rapid growth on manufacturing industries and service sector. Bengaluru led the IT, aerospace, and R&D due to this these areas being called Silicon Valley of India" (Ragothaman, 2024). Chennai often called Detroit of India due to heavy manufacturing in the electronic and automobiles such as Ford, Hyundai, Nokia, and Samsung (Hossain, 2022). Cities like Coimbatore and Madurai also contribute significantly to the region's industrial landscape especially in textile production (Garg, 2019)

4.4 Additional Details

4.4.1 Demographics & Workforce

Mumbai – Pune industrial region employs more than 15 lakh highly skilled workers from nearby IITs. (Muhammed, 2023). Bengaluru - Tamil Nadu have skilled IT and engineering workforce educated from the Indian Institute of Sciences (IISc) in Bengaluru and more than 750 technical institutions in Tamil Nadu (Bengaluru & Chennai: The next big manufacturing destinations, n.d.)

5 Literature Review

5.1 Introduction to Literature Review

5.1.1 Previous Studies

In previous studies of researchers, it states that the spread of industrial regions in India is not random. Establishing industries together in one area is also called industrial cluster (Ketels, 2023) prominent economist Michael Porter studied when companies cluster together they become more successful (Ketels, 2023) by sharing the infrastructure, electricity, etc. (Ketels, 2023)

5.1.2 Importance of These Studies

Developing country like India having creating industrial region will progress the national economy by providing employment of

Opportunity and its positive impact. Creating more employment opportunities and high standard of living (Industrial clusters in India: Mumbai–Pune belt & Bengaluru IT hub (10+ page detailed notes)) (Ketels, 2023)

5.1.3 Important Factor

Researchers found the reasons behind the industrial region such as good infrastructure, skilled labor, areas with better transportation, electricity (Kaur, n.d.) the area that's

have developed through transportation, more skilled labor attracts more companies and industries with government initiatives like national corridor development programme(NICDP) to meet international standards (Department for Promotion of Industry and Internal Trade (DPIIT))

5.2 Industrial Region in India

5.2.1 Concept of Industrial Clustering

To understand how regions evolved, economists depend on the "theory of location of industries by Alfred Weber (Vision IAS, 2024) this theory shows that lowest transportation cost and lowest cost of labor will influence the location of industrial regions (Vision IAS, 2024) in modern time studies follow new theory called "new economic geography" it says when small industries start in an area it will attract more business around that.

5.2.2 Historical Development

Studies show that development happened in three stages such as

- British era – for easy transportation of raw material, more ports were built near the coastal area (Kaur, n.d.)
- Post independence - after India gained independence the government realized that only the port cities were rich so they encouraged new factories to open near existing industrial region
- Economic reforms – India opened up to globalization, which attracted more investment to build more industries especially in western and southern India

5.2.3 Major Industrial Region

Studies focus on industrial region such as Mumbai – Pune, mainly traditional and automobile industries and Bengaluru – Tamil Nadu for IT service and electronic manufacturing (Industrial clusters in India: Mumbai–Pune belt & Bengaluru IT hub (10+ page detailed notes)) research shows the reason behind the success were government policy and cooperation (Tirkey, A comparative analysis: How geographic location and state level policies influence the inflow and impact of FDI in different Indian cities, 2026)**5.3 Studies on Mumbai – Pune Industrial Region**

5.3.1 Development of Industries

Mumbai – Pune industrial region was India's oldest and mixed industrial area (Kaur, n.d.)

Its build on British colonial era because of the port and humid climate, and the opening of Suez Canal gave the title of "Manchester of the east" for this region (Kaur, n.d.). Over time the region became more crowded unable to expand, high land costs this led to shift to the areas like Thane, Navi Mumbai, Ambernath, Pune and Chakan (Kaur, n.d.)

5.3.2 Role of Port and Transportation

Studies show that the main factor behind the success of an industrial region is the transportation and connectivity such as Jawaharlal Nehru port trust (JNPT). This is India's largest container port which allows companies easy access for trade (Industrial clusters in India: Mumbai–Pune belt & Bengaluru IT hub (10+ page detailed notes)) and 94.5-kilometer Mumbai – Pune expressway helps connect the inland transportation and movement of the goods

5.3.3 Economic Growth

Studies show that adapting to new trends helps industries grow instead of sticking to old method and innovation earlier it was traditional textile industries but now there is a wide Variety of mixed industries like;

- Automobile -region like Pimpri, Chinchwad, and Chakan from Pune were dominant for car manufacturing like Tata Motors, Bajaj Auto, Mercedes Benz, Volkswagen (Kaur, n.d.)
- Petrochemical and pharma – because oil was discovered off the coast of Mumbai at Bombay high massive chemical and plastic factories opened here the region is also a leader in pharma manufacturing (Kaur, n.d.)
- Employment – studies note that the industries in this single corridor provide jobs to over 15 lakh people (Kaur, n.d.)

5.3.4 The Problems

Earlier studies also concluded that this incredible growth has come at a very high cost, researchers warn that the Mumbai – Pune industrial region is choking on its own success. The area suffers from problems such as power supply, outdated machinery, high land cost (Kaur, n.d.) rapid urban growth has destroyed the natural vegetation and forest and increasing air pollution and traffic congestion(Kaur, n.d.)

5.4 Studies on Bengaluru – Tamil Nadu Industrial Region

5.4.1 Fast-Growing Corridor

Studies show that Bengaluru – Tamil Nadu industrial belt is one of the fastest growing modern corridors, created after India's independence by the support from the government as the Chennai – Bengaluru Industrial Corridor (CBIC) (Wikipedia contributors, 2026) The region's early industrial base was anchored by public sector unit such as Hindustan Machine Tool (HMT) and Bharat electronics limited(BEL), established by the government in Bengaluru during the post-independence period. The corridor also includes Tumakuru Industrial Township, identified under CBIC master plan as the core to future manufacturing expansion (Japan International Cooperation Agency; PricewaterhouseCoopers Co., Ltd.; Nippon Koei Co., Ltd., 2015)

5.4.2 Growth of IT in Bengaluru

A vast amount of research has been conducted in this cluster, This region also called "Silicon Valley of India" (Goel, 2025) due to the early government investment in the

science and technology. Such government initiatives include aerospace and defense research lab like HAL and government aided technological universities.

5.4.3 Importance of Skilled Labor and Innovation

Bengaluru has highly capable workers who are well educated and proficient in English in engineering field attracting foreign tech companies like Texas instruments, IBM, Microsoft, and Google for their R&D operations (Goel, 2025)

5.4.4 Manufacturing and Automobile in Tamil Nadu

While Bangaluru focuses on software, Tamil Nadu focuses on hardware, studies show that Tamil Nadu mainly focuses on the manufacturing of electronics this region also called "Detroit of India" especially in Chennai companies like Renault Nissan, Ford, and Hyundai manufacture cars (Garg, 2019)

Studies also show a new emerging hub between the Mumbai – industrial region and Bengaluru – Tamil Nadu "Hosur" benefiting from software and hardware from neighbouring industrial belts this region is also becoming a hub for electric vehicles (EV) (United Nations Industrial Development Organization (UNIDO), United Nations Industrial Development Organization)

5.4.5 Role of Government

The studies often point out the involvement of the government, Tamil Nadu and Karnataka governments created "special economic zone" and technological park where companies get benefits through tax cuts, electricity, and high-speed internet (Frost & Sullivan, 2018)

In Mumbai – Pune industrial region researchers warned about rapid population growth, traffic congestion, water crisis in Bangaluru (Hossain, 2022)

5.5 Comparison of These Industrial Regions

5.5.1 Difference in Industrial Regions

- Mumbai – Pune – is a traditional industrial setup. It relies on a sea port, chemical plants, car manufacturing and financial banking system located in Mumbai (Kaur, n.d.)
- Bengaluru – Tamil Nadu – Bengaluru expanded in the economy through human knowledge, IT software, research, and digital innovation (Goel, 2025) meanwhile Chennai and Hosur with manufacturing of heavy engineering, and electronic manufacturing (Goel, 2025)

5.5.2 Role of Infrastructure and Policy

The two regions depended on different types of infrastructure. Mumbai – Pune on physical geography – especially its sea port and early railway built by the British (Kaur, n.d.) Bengaluru – Tamil Nadu region due to government policies that creation

of hundreds of engineering colleges and the building of dedicated IT software parks (Goel, 2025) both regions make a vital economic contribution to India they are the core to the country 's exports, attract foreign direct investment, and provide employment opportunities (Department for Promotion of Industry and Internal Trade (DPIIT))

5.5.3 Research Gap

Even there is thousands of research papers have been written about this industrial region there is still major research gaps exist

- Studied in separately – most studies focus only one region at a time, they write whole paper on Bengaluru IT sector or Pune automobiles, very few studies put these two multi-corridors next to each other to compare.
- Lack of environmental comparison –there is not enough research comparing how industries damage the environment.

5.5.4 Why It Is Important

This study is very important for its bridge these exact gaps; by directly comparing an older, traditionally expanded, heavy manufacturing industry with new modern technology and skill-driven work force, this paper will provide a complete idea of how different industries influence a region.

Understanding how a "factory economy" and "knowledge economy" on the basis of employment creation, local infrastructure and city congestion will give important information to the government authority then it will help planners to design better more sustainable cities in the future without destroying the local environment

6. Analysis

6.1 Industrial Structure Analysis

Sector	Mumbai – Pune industrial region	Bengaluru – Tamil Nadu industrial region
Automobile	Tata Motors, Bajaj Auto, Mercedes Benz, Volkswagen	Hyundai, Renault – Nissan, Ashok Leyland
IT & Software	Pune IT parks and service sector	Bengaluru India's largest IT hub
Petrochemical	Refineries and chemical industries near Mumbai	Limited petro chemical industries
Electronics	Small manufacturing clusters	Samsung, Foxconn, Nokia
Aerospace	Minor presence	HAL, ISRO, DRDO, R&D, centers
Textile	Historic cotton textile industry	Textile hubs in Coimbatore

Figure 6.1 industrial structure comparison of Mumbai – Pune and Bengaluru – Tamil Nadu region

6.1.1 Mumbai- Pune

- Automobiles – Pune is a leading global manufacturer of motor vehicle such as Tata Motors, Bajaj Auto, Mercedes Benz, Volkswagen. This region contributes to produce 35 % of India's automobile output.
- Petrochemicals – Due to the Mumbai high oil field, the region hosts major refineries operated by Reliance, BPCL, and HPCL
- Engineering and textiles – known as the "Manchester of the east" Mumbai started to grow with its base on cotton textiles while Pune expanded by including heavy engineering.

6.1.2 Bengaluru – Tamil Nadu region

- IT and electronics – Bengaluru also known as the "Silicon Valley of India" leading software exports, IT services, and startups. Chennai is a major hardware and electronics manufacturing cluster. International companies such as Texas Instruments, IBM, Microsoft, and Google have established major R&D operation here.
- Automobile and EV – Chennai is well known as the "Detroit of Asia" due to its mass production of automobile in nearby Hosur has rapidly evolved as a leading electric vehicle manufacturing, housing companies, like OLA Electric, Ather energy, and TVS.
- Aerospace – Bengaluru serves India's aerospace research and manufacturing supported by the HAL and ISRO, Tamil Nadu developing components to aerospace
- Textile production – cities like Coimbatore and Madurai is leading in the production of the textile in south India.

6.1.3 Key Differences

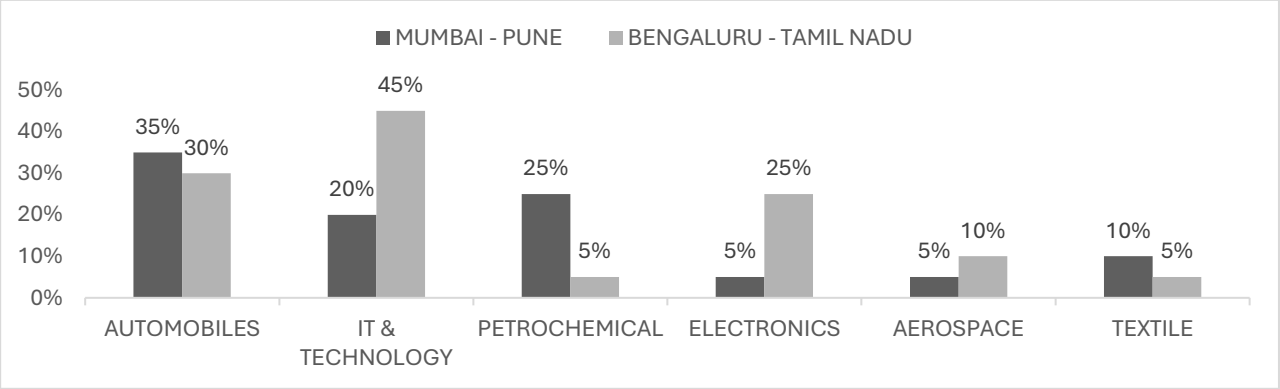

Figure 6.1.3 industrial structure comparison of Mumbai – Pune and Bengaluru – Tamil Nadu industrial region

Mumbai-Pune's and its legacy of coastal trade, heavy industries, and chemical produce, in other hand, the Bengaluru-Tamil Nadu region is led by skilled educated workforce, focusing on latest software, R&D and advanced precision engineering.

6.2 Growth and Development

6.2.1 Mumbai – Pune region

Grew in different time era starts with the cotton textile era (1850 – 1950) due to British port access and Suez Canal. It later expanded into engineering (1950 – 1990) and experienced a sudden change of foreign direct investment in automobile and IT post 1990.As Mumbai became congested, industrial move toward tier2 areas like Thane, Navi Mumbai, Ambernath, and Chakan. Surrounding districts such as Nashik, Ahmednagar, and Sangli also gradually expanded industrial development (Industrial clusters in India: Mumbai–Pune belt & Bengaluru IT hub (10+ page detailed notes))

6.2.2 Bengaluru – Tamil Nadu

initially led by heavy public sector units like HMT, and Bharat Electronics, the region saw rapid growth in the 1990s, due to establishment of software technology.

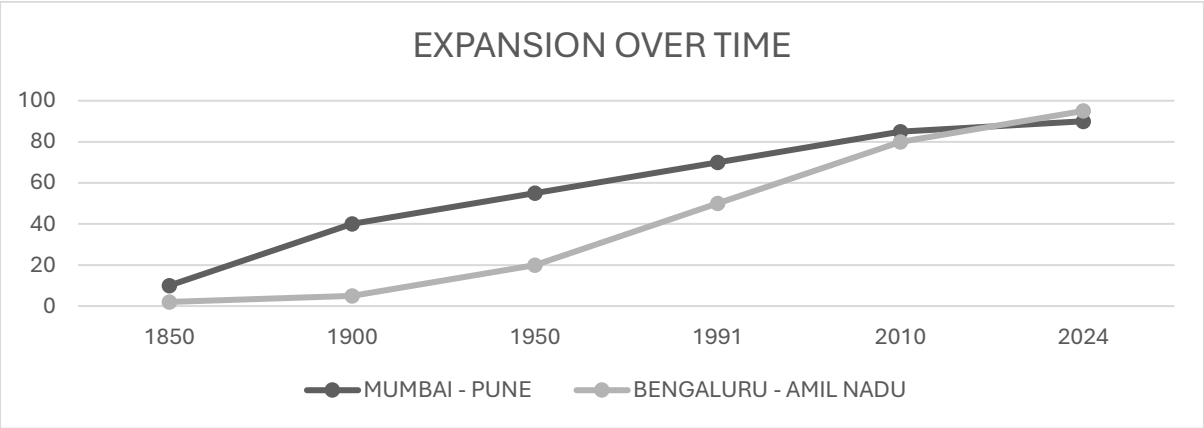

Figure 6.2.2 industrial expansion over time in Mumbai – Pune and Bengaluru Tamil Nadu region

6.3 Infrastructure and Location

6.3.1 Transportation Network

Mumbai-Pune's developed by the coastal port access and Mumbai-Pune Expressway. Bengaluru - Tamil Nadu region use southern seaports (Chennai, Ennore, Tuticorin, Mangalore) and is being connected to the Chennai - Bengaluru Industrial Corridor (CBIC).

6.3.2 Raw Material

Mumbai used its vast cotton-producing industries and near to the Mumbai high petroleum field in other hand, Bengaluru - Tamil Nadu lack of coal reserves and shifting from agricultural to a heavy reliance on skilled human capital.

6.3.3 Energy Supply

Both regions have lack of coal and have to depend on the hydroelectric power for energy. Mumbai-Pune depends on Western Ghat hydroelectric power and Bengaluru - Tamil Nadu depends on dams like Pykara, Mettur, and Sivasamudhram.

6.3.4 Government Policies

Maharashtra successfully established extensive industrial estates, the southern region boom was fuelled by Karnataka government's progressive IT policies and Tamil Nadu's proactive development of special economic zones.

6.4 Employment and Economic Opportunities

6.4.1 Employment Generating

Tamil Nadu ranks one of the largest number of industries and workers led by tech and service sector Bengaluru registers India's highest employment (7.2%) and productivity (8.1%) growth rate the Mumbai-Pune region workers over 15 lakh people in its manufacturing sectors and special hubs like Hinjewadi alone created 4 lakh IT professionals. This skilled workforce IITs in the Mumbai-Pune region and IISc along with over 750 technical institutions in Tamil Nadu.

6.4.2 Urbanization and Migration

Initially Mumbai industrial region carried out with migration workers from nearby states by providing employment opportunities. Bengaluru's modern IT sector attracted skilled workers, leading rapid urban development in the region.

6.4.3 Regional Economic Development

Both regions contribute to the Indian economy, but high concentration of industries has led to high land cost, water scarcity, congestion, pollution, driving a shift of industries toward tier-2 and tier-3 cities.

7 Conclusion

This study compares India's two major industrial regions Mumbai-Pune and Bengaluru-Tamil Nadu industrial region, how they developed, how they influence regional development and the Indian economy. Mumbai-Pune region developed due to the coastal advantage, port access and connectivity, with industries such as textiles, petrochemicals, engineering and automobiles. Bengaluru - Tamil Nadu expanded rapidly after Indian independence, this region attracted more skilled workforce, which led to more industrial clusters, IT companies, both these regions influenced in generating employment, attracting foreign investments, Mumbai – Pune attracting factory workers and Bengaluru – Tamil Nadu taking skilled professionals from across India, both regions different from each other Mumbai-Pune was focused on heavy manufacturing and Bengaluru - Tamil Nadu focused on software and modern technological advancements. By a comparative study of these two industrial regions, this paper shows how

geographical location, infrastructure, government policy and technological advancement impact on regional development, and due to some limitation its shifting to new industrial growth toward tier-2 cities like Hosur and Tumakuru.

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